

Programming and Fundamental Algorithms

WM908-15 — revision saved on 28th May 2026 at 08:30

Deadline: 31st May 2026

Approved on 27th May 2026 at 12:07.

Basic module details

Month of introduction/change

October

Year of introduction/change

2026

Module title

Programming and Fundamental Algorithms

Module duration

4 weeks

Department for SITS

WMG

Summary of changes or other notes for reviewers

28/04/2026

The module is technical and a 100% assessment suits delivery across 4 weeks. Positive module reviews and PTES feedback suggest this approach is working well, so no need to change. Agreed with Course Leader and AQ.

Rationale for introduction, revision or discontinuation

Replacing WM908 with 15 CATS alternative to support SCAV, SAE (tbc) and CSM

Affected departments

- WMG

Module leader

Kurt Debattista (0673615)

Level

7

Level for SITS

M1

Credit value

15

Module delivery

Blended

Study locations

Location	Primary
University of Warwick Main Campus, Coventry	Yes

Module description

Introductory description

Current progress in information technology has meant the majority of organisations are using IT to innovate, and without an understanding of fundamental computing concepts future managers, scientists, engineers will be unable to take strategic decisions and provide critical thinking on most projects in full confidence.

The module is focused around two core themes in computing: algorithms and programming/development. These two concepts go hand in hand and it is understood that to be a good developer, algorithmic concepts need to be comprehensively understood and students should be able to critical apply knowledge from the theoretical aspects towards practical implementations of solutions for complex system designs in business, engineering, science and IT.

The programming focus is based on a high level programming language such as the C/C++ programming language and/or python, considered by many as one of the more useful programming languages and still the language of choice in many industries – once mastered the transition to the other popular programming languages Java, C-sharp, Objective C can be relatively straightforward. Importantly, this is not about teaching programming but about forming a conceptual understanding of computing principles with programming as a vehicle to further grasp these concepts with the added bonus of adding an important skill to the CV. Future editions may adopt other programming languages.

Principal module aims

The over-arching aim of this module is to give students the ability and confidence to solve problems efficiently using computers. The sub-aims of this module are:

- Form an understanding of some classic algorithms from the literature
- Develop the understanding of which solutions/algorithmic paradigms work best for certain types of problems
- Design straightforward algorithms for yet unseen problems that have straightforward solutions
- Learn programming methods and how to design good code for a proposed algorithm

Learning outcomes

- Master a sound, conceptual understanding of the theory and concepts of programming and fundamental algorithms and data structures. [AHEP:4, M5,M4] [CITP 3.1.1]
- Program effectively and independently in a high-level programming language at an intermediate to advanced level. [AHEP:4, M13] [CITP2.2.1]
- Take, straightforward to complex, algorithmic concepts, and implement them correctly. [AHEP:4, M2,M4] [CITP2.2.4]
- Autonomously distinguish the right solution for a given problem from amongst a set of algorithmic and programming tools. [AHEP:4, M3] [CITP3.2.2]

Outline syllabus for publication

Introduction Sorting as an introduction to algorithms Data structures Complexity and decision making Brute force and divide and conquer methods for solving problems Dynamic programming and greedy methods Exhaustive search and recursion Advanced Data structures Misc. algorithms (eg string matching, spatial data structures etc.) as part of in-class tutorials, introduced throughout the module. Limitations of algorithms and coping with limitations Conclusions, recap and next steps

Tutorials Introduction to programming Introduction to Types and Commands Dealing with pointers Generics and abstract data types Concepts of Object Oriented Programming Inheritance, Polymorphism File I/O Standard Template Library Modern C++

Supplementary detail on syllabus and teaching strategies to inform the approval (not for publication)

(this was explaining the differences from prior running; module is now running so have removed them)

Should this module have a reading list?

No

Module delivery

Minimum student number

20

Maximum student number

90

Estimated student number

30

Teaching split

Department	Percentage
WMG	100%

Skills, competencies, and attributes

Subject specific skills

Mathematical skills, programming skills

Transferable skills

Technology literacy, adaptability

Includes research element

Yes

Research element description

Many research students have attended the 10 CATS version of this course before (and particularly in its previous incarnation as CY903) - it forms a solid foundation for students working in the mathematical sciences to learn how to program and run code. Many of the in class examples are, usually, simulation based.

Interdisciplinary

Yes

Interdisciplinary description

The examples shown in terms of programming will cover a number of areas around science and engineering, broadly falling in the mathematical sciences domain.

International

No

Learning hours breakdown

Lecture

Required: 15 × 1 hour Optional: 0 × 0 hours

Seminar

Required: 0 × 1 hour Optional: 0 × 0 hours

Tutorial

Required: 15 × 1 hour Optional: 0 × 0 hours

Project Supervision

Required: 0 × 0 hours Optional: 0 × 0 hours

Demonstration

Required: 0 × 0 hours Optional: 0 × 0 hours

Practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Supervised practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Fieldwork

Required: 0 × 0 hours Optional: 0 × 0 hours

External visit

Required: 0 × 0 hours Optional: 0 × 0 hours

Work based learning

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (scheduled sessions)

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (independent)

Required: 11 × 1 hour Optional: 0 × 0 hours

Autonomous skills development (Degree apprenticeship provision only)

Required: 0 × 0 hours Optional: 0 × 0 hours

Other activity

Required: 29 × 1 hour Optional: 0 × 0 hours

Details of other activity

29 hours of student self directed study. Guidance on self directed study will be provided in class.

Placement

0 hours

Private study

20 hours

Details of private study and independent learning

Private study will involve the understanding of programming concepts and algorithms.

Assessment

60 hours

Total study hours for module

150 hours

Assessment

Reassessment model

Individual

Assessment components

Title	Type	Length	Weight	Final chronological	Eligible for self-certification
Games Programming Assignment	Project - Individual	4000 words	100%	Yes	Yes (extension)
Student preparation and completion time (hours)	60				
Description	This is the module assignment. It is an original programming application with a brief report outlining motivation and solutions.				
Reassessment component					
Title	Type		Length		Eligible for self-certification
Games Programming Assignment	Project - Individual		4000 words		No
Description	A game application accompanied by the source code and a brief report outlining the motivation, methodology, and proposed solutions.				

Must the student pass all parts of the assessment to pass the module?

Yes

Methods for providing feedback on assessment

The assignment will be marked and feedback provided. Feedback will also be provided in class via the tutor demonstrating a solution to similar problems.

Assessment learning outcomes

	Games Programming Assignment (100%)
Master a sound, conceptual understanding of the theory and concepts of programming and fundamental algorithms and data structures. [AHEP:4, M5,M4] [CITP 3.1.1]	X
Program effectively and independently in a high-level programming language at an intermediate to advanced level. [AHEP:4, M13] [CITP2.2.1]	X
Take, straightforward to complex, algorithmic concepts, and implement them correctly. [AHEP:4, M2,M4] [CITP2.2.4]	X
Autonomously distinguish the right solution for a given problem from amongst a set of algorithmic and programming tools. [AHEP:4, M3] [CITP3.2.2]	X

Availability

Availability of module

Course	Route	Year of study	Type of availability	Required
Postgraduate Taught Smart, Connected and Autonomous Vehicles (Full-time) (TWMS-H1SE)		1	Optional Core	No
Postgraduate Taught Cyber Security Management (Full-time) (TWMS-H1SH)		1	Optional	No
Postgraduate Taught Games Engineering (Full-Time) (TWMS-H1SK)		1	Core	No
Postgraduate Taught Smart, Connected and Autonomous Vehicles (Part-time) (TWMS-H1TE)		1	Optional	No
Postgraduate Taught Engineering Competence (Smart, Connected and Autonomous Vehicles) (Degree Apprenticeship) (EWMS-H1U2)		1	Optional	No

Has pre-requisite modules?

No

Has post-requisite modules?

No

Has anti-requisite modules?

No

Any additional resource requirements

None.